

Shubham Kumar Mitra

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Summary

Computer Science undergraduate (B.Tech) with strong foundations in Data Structures, Algorithms, and Object-Oriented Programming. Specialized in AI/ML, computer vision, and scalable backend systems. Built 4+ real-time, data-driven applications integrating machine learning models with production-ready architectures. Seeking internship opportunities to apply analytical thinking and engineering skills to real-world industrial challenges.

Education

Vellore Institute of Technology, Chennai	2022 – Present
Bachelor of Technology in Computer Science Engineering	CGPA: 8.77
Hartmann College, Uttar Pradesh	2022
Class XII (PCM)	85%

Technical Skills

Languages: C++, Python, JavaScript, SQL

Frameworks & Libraries: Next.js, React, FastAPI, OpenCV, MediaPipe, Scikit-learn (Supervised ML), Pandas, NumPy

Databases: PostgreSQL, MongoDB

Core CS: Data Structures & Algorithms, OOP, DBMS, System Design Basics

Tools & Platforms: Git, GitHub, Linux, REST APIs, Vercel

Professional Exposure

Technology Executive Intern May 2026 – Jul 2026
Seatrick Innovations Pvt Ltd, Singapore-based Maritime Technology Company

- Supported testing, analysis, technical coordination, and project execution for digital initiatives serving the maritime technology ecosystem.
- Assisted in workflow documentation and operational assessment activities within an international, Singapore-based technology environment.

Summer Research Intern 2026
Vellore Institute of Technology

- Engaged in research-oriented technical work involving literature study, experimentation, analysis, and documentation.
- Strengthened analytical thinking and problem-solving through structured academic research workflows.

Experience

AI/ML Developer (Personal Projects) 2025 – Present
Independent Development, Remote

- Built 5+ AI and computer vision systems using Python, OpenCV, and MediaPipe enabling real-time processing at 25+ FPS.
- Architected RESTful APIs using FastAPI and MongoDB supporting scalable backend workflows handling 1000+ records.
- Constructed machine learning pipelines for NLP and pose classification tasks, improving processing efficiency by 30%.

Projects

Sports Companion Live
Next.js, TypeScript, PostgreSQL, ML Models

- Built a full-stack analytics platform delivering insights on workouts and performance metrics.
- Developed a hybrid prediction engine combining heuristic logic with deep learning achieving 80%+ accuracy.
- Integrated computer vision-based height estimation using spatial landmark detection at 20+ FPS.
- Structured a PostgreSQL schema with 5 tables, 10+ attributes, and 4 foreign-key constraints ensuring normalized storage.

AI Badminton Engine

Demo

Python, OpenCV, MediaPipe

- Engineered a real-time pose tracking system processing 30+ skeletal keypoints per frame.
- Achieved stable performance of 25+ FPS for smooth gameplay analysis.
- Designed a finite-state system with 5 states ensuring deterministic and consistent gameplay transitions.
- Reduced inference latency by 35% through optimized frame processing.

Breathe Map

Live

Next.js, PostgreSQL, Scikit-learn

- Built a city-scale AQI analytics platform with interactive dashboards and real-time data visualization.
- Managed 5 relational tables with Row Level Security ensuring safe multi-user access.
- Delivered 6 REST API endpoints for AQI prediction and analytics workflows.
- Improved API response efficiency by 25% through query optimization.

Problem Solving

- Solved 100+ DSA problems across platforms such as LeetCode focusing on arrays, trees, graphs, and dynamic programming.
- Strengthened skills in time-space optimization and scalable algorithm design.

Certifications

IBM Agile Development & Scrum Certification

Verification: coursera.org/verify/9N2HZHLJCHXX

Exploring Computer Networks (Cisco Packet Tracer)

Figma UI/UX Design Certification

Credential ID: UC-1bab30ee-bc80-448b-8104-32595a8e42c6

Achievements & Activities

- Built 4+ production-grade projects integrating AI, backend systems, and analytics pipelines.
- Continuously exploring machine learning, computer vision, and scalable system design.
- Actively practicing algorithmic problem solving to enhance technical interview readiness.